

Credit Card Fraud Detection Using Machine Learning

(Project Report)

1. Introduction

Credit card fraud is one of the major challenges faced by financial institutions. With the increase in online transactions and digital payments, fraudulent activities have also increased significantly. Fraud detection systems are used to identify suspicious transactions and prevent financial loss.

Machine learning techniques are widely used for fraud detection because they can analyze large volumes of transaction data and detect unusual patterns. The goal of this project is to analyze credit card transaction data and identify fraudulent transactions using machine learning techniques.

2. Problem Statement

- Credit card fraud occurs when someone uses another person's credit card or account information to make unauthorized purchases or access funds. It is a significant problem for banks, businesses, and customers, leading to financial losses and reputational damage. Fraud can happen both online and in physical stores, making it difficult to track and prevent.
- The main challenge is to identify fraudulent transactions accurately among a large number of legitimate transactions. Traditional methods often fail to detect new or sophisticated fraud patterns and may generate many false alarms, which increases operational costs and reduces efficiency.
- The goal of this project is to develop an intelligent fraud detection system that can automatically detect suspicious transactions in real time. The system should maximize detection of fraudulent activities while minimizing incorrect alerts, helping financial institutions reduce losses and protect customers from fraud.

Objectives

1. Detect Fraudulent Transactions: Build a system that can accurately identify credit card transactions that are fraudulent.
2. Minimize False Positives: Reduce the number of legitimate transactions incorrectly classified as fraud to avoid inconvenience to customers.
3. Analyze Transaction Patterns: Explore transaction characteristics such as amount and timing to understand trends in fraudulent activity.

4. **Develop a Predictive Model:** Use machine learning techniques to predict the likelihood of a transaction being fraudulent based on historical data.
 5. **Enhance Financial Security:** Provide a solution that helps financial institutions and businesses minimize financial losses and protect customers.
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3. Dataset Overview

The dataset used for this project contains credit card transaction records with multiple features representing transaction characteristics.

According to the dataset inspection:

- Total records: **284,807 transactions**
- Total features: **31 columns**
- Target variable: **Class**
 - **0 → Normal transaction**
 - **1 → Fraudulent transaction**

Most of the variables are transformed features generated using **Principal Component Analysis (PCA)** for security purposes.

Key features include:

- **Time** – Time elapsed between transactions
 - **Amount** – Transaction amount
 - **V1 to V28** – PCA transformed features
 - **Class** – Target variable indicating fraud or non-fraud
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4. Data Inspection

Before building a machine learning model, the dataset was inspected to understand its structure and characteristics.

Several inspection functions were used:

- `df.head()` – Display the first few rows of the dataset
- `df.tail()` – Display the last few rows

- `df.info()` – Provides information about data types and non-null values
- `df.describe()` – Displays statistical summary of the dataset
- `df.shape` – Shows the number of rows and columns
- `df.columns` – Lists column names
- `df.dtypes` – Shows the data type of each column

From the inspection results:

- All columns contain **numeric values**
- Most variables are **float64**
- The target variable **Class** is integer type

The dataset contains **284,807 rows and 31 columns**, indicating a large dataset suitable for machine learning analysis.

5. Missing Value Analysis

Missing values can negatively impact machine learning models.

Using the following command:

```
df.isnull().sum()
```

The result showed:

- **No missing values in any column**

This means the dataset is already clean and ready for analysis.

6. Duplicate Data Handling

Duplicate records were checked using:

```
df.duplicated().sum()
```

Result:

- **1081 duplicate rows were found**

These duplicates were removed using:

```
df.drop_duplicates()
```

Removing duplicate records improves data quality and prevents bias in the machine learning model.

7. Target Variable Analysis

The target variable **Class** represents whether a transaction is fraudulent.

Value distribution:

Class Meaning

- 0 Normal transaction
- 1 Fraud transaction

After removing duplicates:

- Normal transactions: **283,253**
- Fraud transactions: **473**

Fraud transactions represent only **0.16% of the dataset**, which means the dataset is **highly imbalanced**.

This imbalance makes fraud detection a challenging machine learning problem.

8. Skewness Analysis

Skewness measures the asymmetry of data distribution.

Interpretation:

- **0 → symmetric distribution**
- **Positive value → right-skewed distribution**
- **Negative value → left-skewed distribution**

Some variables show high skewness values, indicating the presence of extreme values and non-normal distributions in the dataset.

For example:

- **Amount feature shows high positive skewness**
- **Some PCA variables show negative skewness**

This suggests the presence of outliers in the dataset.

9. Exploratory Data Analysis (EDA)

EDA helps in understanding patterns, trends, and relationships in the data.

Several visualization techniques were used.

9.1 Fraud vs Non-Fraud Transactions

A **count plot** was used to visualize the number of fraudulent and non-fraudulent transactions.

Observation:

- Normal transactions are extremely high
- Fraud transactions are very few

This confirms that the dataset is **highly imbalanced**.

9.2 Transaction Amount Distribution

A histogram was plotted for transaction amounts.

Observation:

- Most transactions have **small amounts**
- Only a few transactions have very high values

This indicates that **high-value transactions are rare**.

9.3 Transaction Time Distribution

A histogram was plotted for transaction time.

Observation:

- Transactions occur throughout the dataset time period
- The distribution shows peaks at certain time intervals

This suggests varying transaction activity over time.

9.4 Transaction Amount by Class

A **boxplot** was used to compare transaction amounts for fraud and non-fraud transactions.

Observation:

- Fraud transactions generally have **smaller transaction amounts**
 - Some normal transactions show very high values
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9.5 Feature Distribution

Histograms were plotted for **PCA features V1–V28**.

Observation:

- Most features follow a **bell-shaped distribution**
- Some features show skewness or extreme values

These features help the model identify hidden patterns in transactions.

9.6 Correlation Heatmap

A correlation heatmap was used to analyze relationships between variables.

Interpretation:

- **Red color** → **strong positive correlation**
- **Blue color** → **negative correlation**
- **Light color** → **weak correlation**

Most features show **low correlation**, which indicates independent variables.

This is beneficial for machine learning models.

9.7 Outlier Detection

Boxplots were used to identify outliers in transaction amounts.

Observation:

- Several extreme values exist in the **Amount** feature
 - These values may represent unusually large transactions.
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9.8 Transaction Time vs Amount

A scatter plot was used to visualize the relationship between **transaction time** and **transaction amount**.

Observation:

- Most transactions occur at lower amounts
- Fraud transactions are scattered among normal transactions

This shows that fraud detection requires analyzing complex patterns rather than simple thresholds.

10. Machine Learning Model

A **Random Forest Classifier** was used to build the fraud detection model.

Random Forest is suitable because:

- It handles large datasets effectively
 - It works well with complex feature interactions
 - It provides feature importance information
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11. Model Evaluation

The model performance was evaluated using classification metrics such as:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1-Score

The confusion matrix helps in understanding:

- Correct fraud detection
 - Missed fraud cases
 - False alarms
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12. Conclusion

In this project, a credit card fraud detection system was developed using machine learning techniques. The dataset was inspected, cleaned, and analyzed using exploratory data analysis methods.

The Random Forest model was used to classify transactions as fraudulent or legitimate. Various visualizations helped understand transaction behavior and fraud patterns.

Due to the highly imbalanced nature of the dataset, fraud detection remains a challenging problem. However, machine learning models can significantly improve the ability to identify fraudulent transactions and reduce financial losses.
